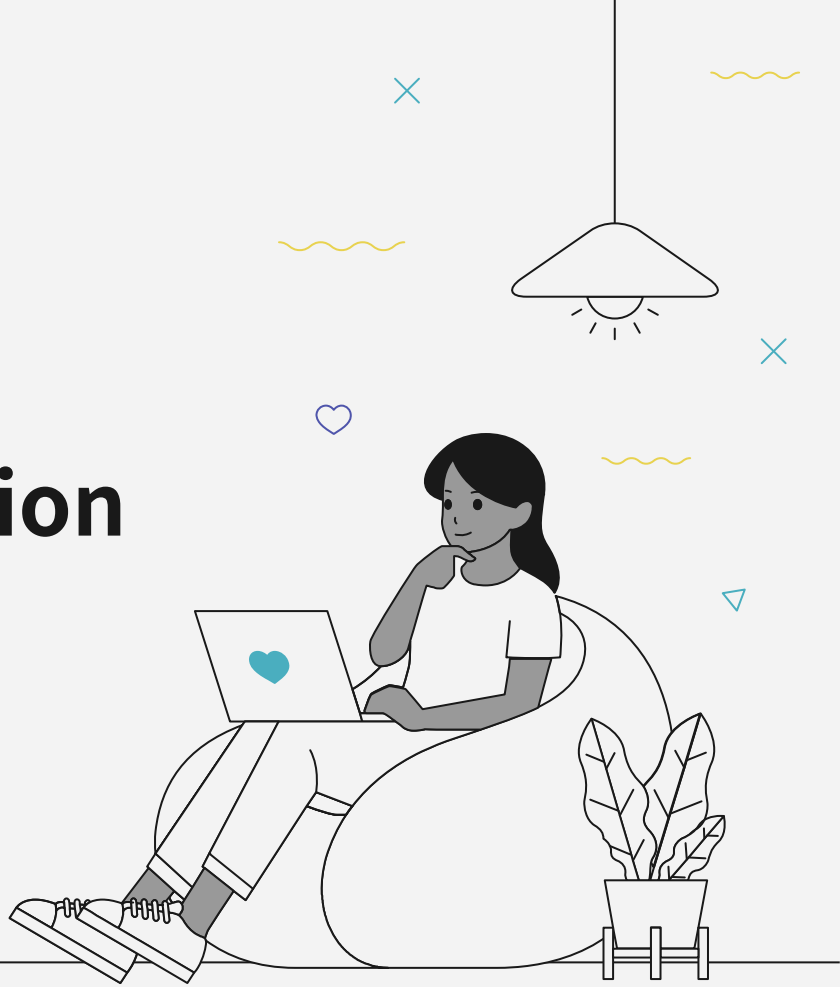
DL Lab 3

Semi-Supervised Intel Image Classification



TAs

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李俊彥 jimmyli90719.ai14@nycu.edu.tw

Training Workflow

1. Supervised Training

Train a CNN with labeled data.

2. Pseudo-Labeling

Predict unlabeled data and keep high-confidence predictions.

3. Self-Training

Train with labeled and pseudo-labeled data.

4. Test Prediction

Select the best validation checkpoint and generate result.csv.

Dataset



buildings



forest



mountain



glacier



sea

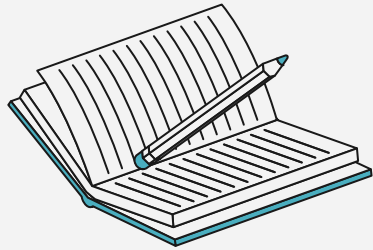


street

Assignment Rules

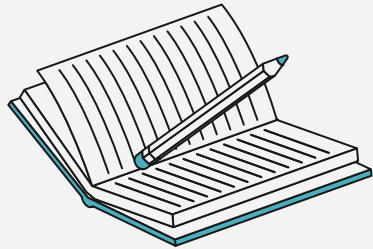
- Complete this assignment individually using PyTorch.
- Implement your own CNN and train it from scratch.
- Do not use pre-built models, pre-trained weights, or external prediction services.
- Use only the provided data. Do not look up or manually label test image.

Rules may be updated or clarified.
Any changes will be announced on the E3.



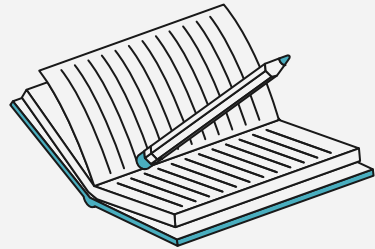
Submission Rules

- Plagiarism results in a score of zero. Do not share solutions, weights, or predictions.
- Incorrect submission format: assignment score $\times$ 0.9.
- Late submissions: assignment score $\times$ 0.9 for each day late.
- Incomplete required work will receive deductions.



Using Kaggle

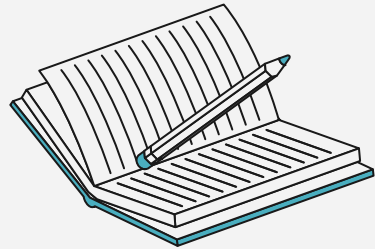
- Join the competition and rename your team to your student ID.
- Download the assignment dataset from the Data tab.
- You may submit up to 5 times per day.



Submission Requirements

Deadline: 10/18 23:59

Notebook	HW3_studentID.ipynb	completed code and training outputs
Report	HW3_studentID_report.pdf	maximum 3 pages
Predictions	result.csv	submit to Kaggle



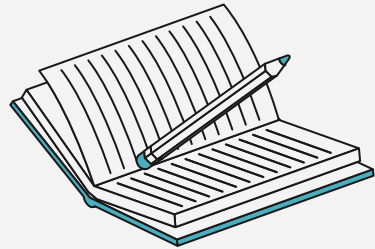
Grading

Code & Model Performance — 70%

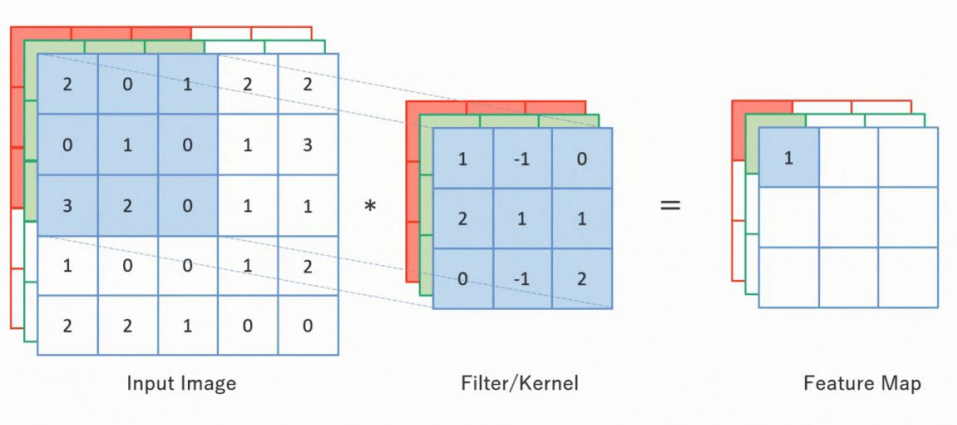
- The TAs will assess your code and model ranking.

Report — 30%

- Explain your experimental design, comparisons, and findings.
- Compare three activation functions.



Convolution in CNNs



$$\text{OutSize} = \left\lfloor \frac{\text{InSize} + 2P - K}{S} \right\rfloor + 1$$

$$\text{OutSize} = \frac{5 + 2(0) - 3}{1} + 1 = 3$$

```
from torch.nn.modules import Conv2d

cnn = Conv2d(3, 64, kernel_size=3, stride=1, padding=1)

print(cnn.weight.shape)
```

[25] ✓ 0.0s

```
torch.Size([64, 3, 3, 3])
```

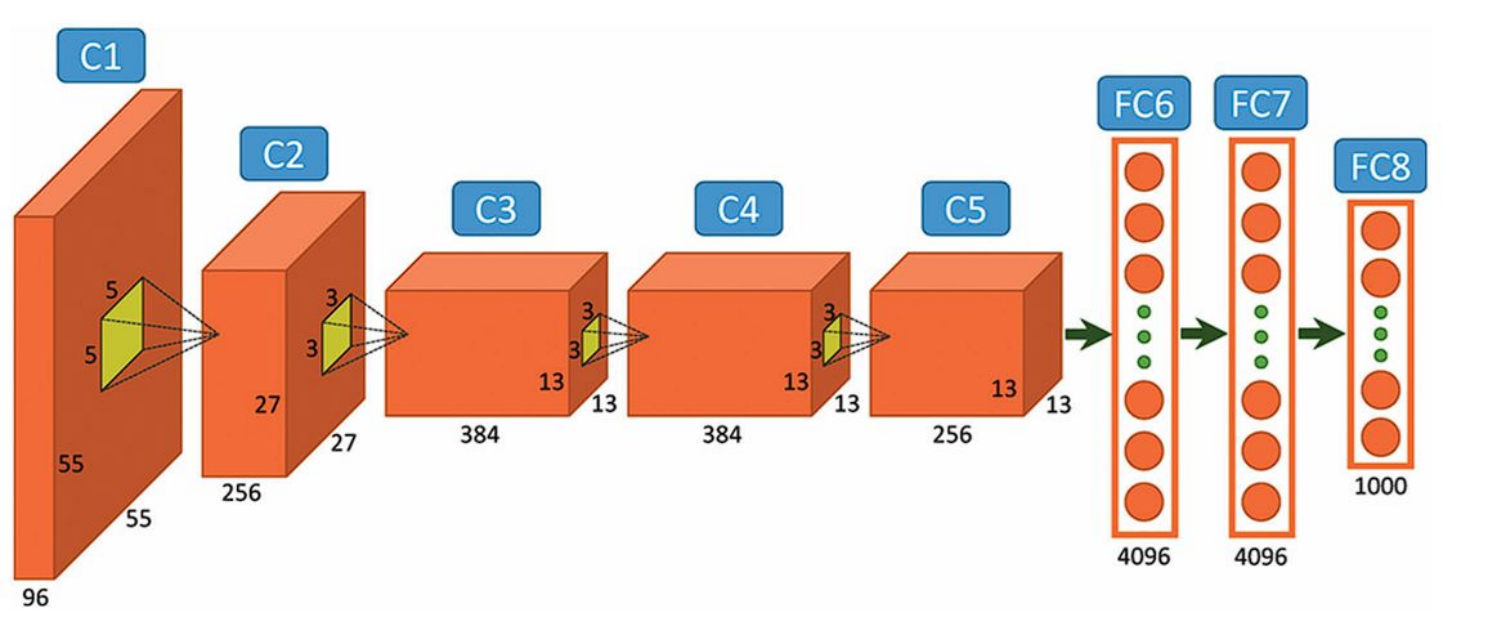
【機器學習 2021】卷積神經網路 (Convolutional Neural Networks, CNN)

Building Blocks

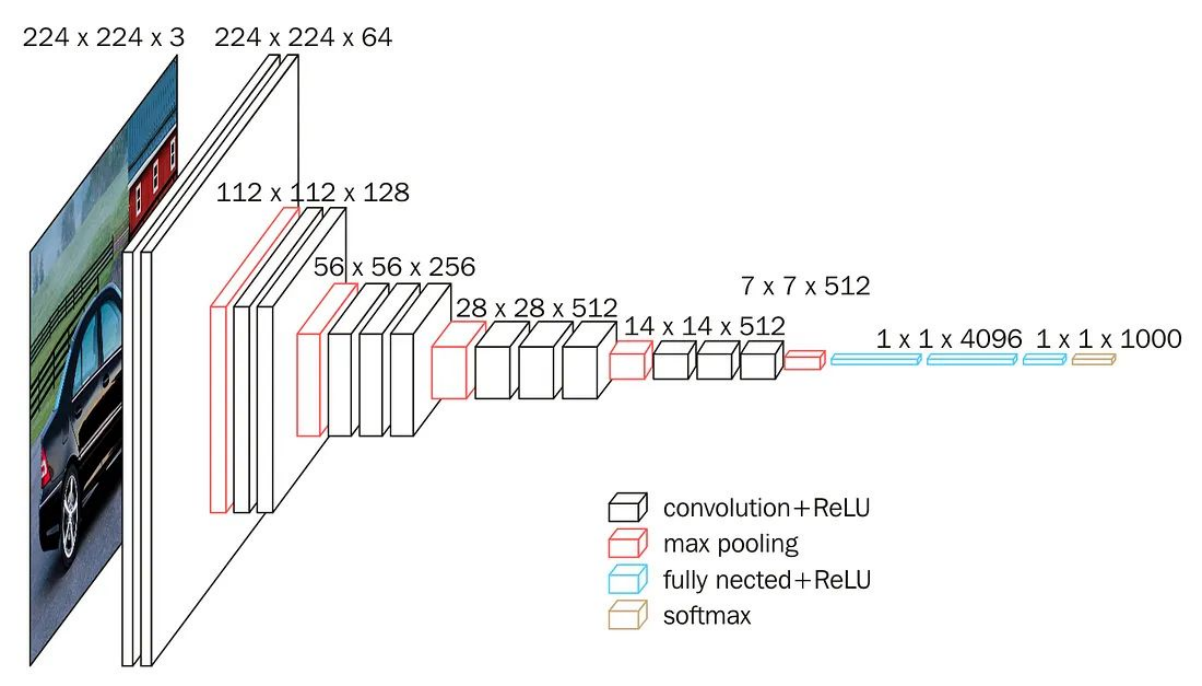
Category	PyTorch Modules	Purpose
Feature Transformation	Conv2d, Linear	Learn and transform features.
Pooling	MaxPool2d, AvgPool2d	Aggregate spatial information and reduce spatial size.
Normalization	BatchNorm2d, LayerNorm	Normalize activations to help stabilize training.
Regularization	Dropout	Reduce overfitting.
Activation	ReLU, LeakyReLU, GELU	Introduce nonlinearity.

[Pytorch basic building blocks]

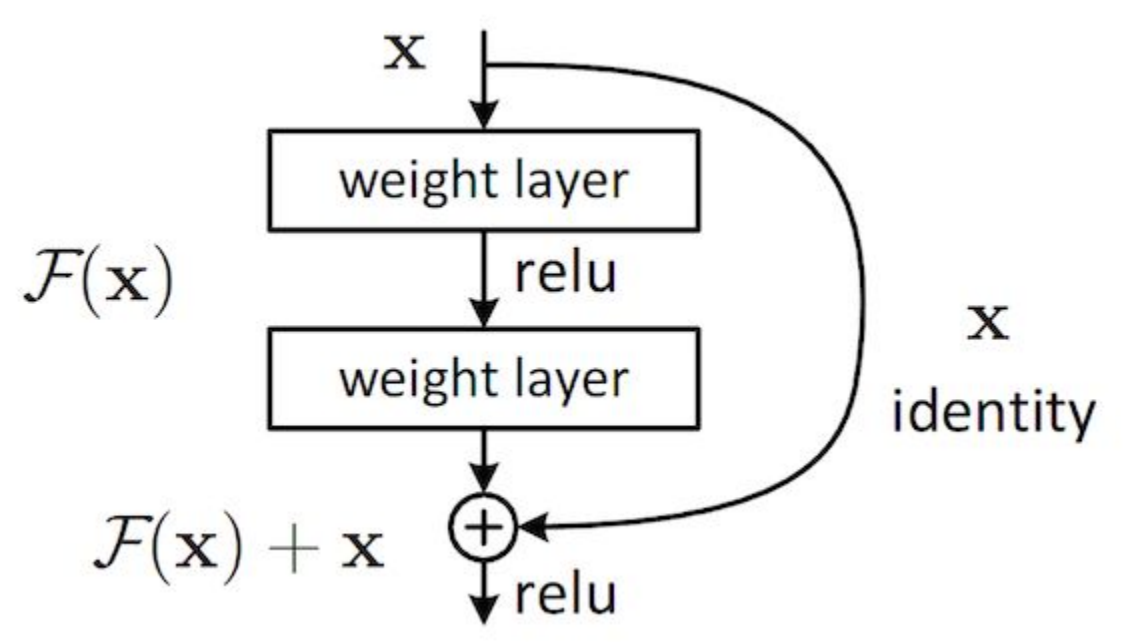
AlexNet（2012）



VGG（2014）


224 x 224 x 3
224 x 224 x 64
112 x 112 x 128
56 x 56 x 256
28 x 28 x 512
14 x 14 x 512
7 x 7 x 512
1 x 1 x 4096
1 x 1 x 1000
convolution+ReLU
max pooling
fully nected+ReLU
softmax

ResNet（2015）


x
weight layer
$\mathcal{F}(\mathbf{x})$
relu
weight layer
x
identity
$\mathcal{F}(\mathbf{x}) + \mathbf{x}$
relu

ConvNeXt（2022）

